

17. Chi Square

A. Chi Square Distribution

B. One-Way Tables

C. Contingency Tables

D. Exercises

Chi Square is a distribution that has proven to be particularly useful in statistics. The first section describes the basics of this distribution. The following two sections cover the most common statistical tests that make use of the Chi Square distribution. The section “One-Way Tables” shows how to use the Chi Square distribution to test the difference between theoretically expected and observed frequencies. The section “Contingency Tables” shows how to use Chi Square to test the association between two nominal variables. This use of Chi Square is so common that it is often referred to as the “Chi Square Test.”

Chi Square Distribution

by David M. Lane

Prerequisites

- Chapter 1: Distributions
- Chapter 7: Standard Normal Distribution
- Chapter 10: Degrees of Freedom

Learning Objectives

1. Define the Chi Square distribution in terms of squared normal deviates
2. Describe how the shape of the Chi Square distribution changes as its degrees of freedom increase

A *standard normal deviate* is a random sample from the standard normal distribution. The Chi Square distribution is the distribution of the sum of squared standard normal deviates. The *degrees of freedom* of the distribution is equal to the number of standard normal deviates being summed. Therefore, Chi Square with one degree of freedom, written as $\chi^2(1)$, is simply the distribution of a single normal deviate squared. The area of a Chi Square distribution below 4 is the same as the area of a standard normal distribution below 2, since 4 is 2^2 .

Consider the following problem: you sample two scores from a standard normal distribution, square each score, and sum the squares. What is the probability that the sum of these two squares will be six or higher? Since two scores are sampled, the answer can be found using the Chi Square distribution with two degrees of freedom. A Chi Square calculator can be used to find that the probability of a Chi Square (with 2 df) being six or higher is 0.050.

The mean of a Chi Square distribution is its degrees of freedom. Chi Square distributions are positively skewed, with the degree of skew decreasing with increasing degrees of freedom. As the degrees of freedom increases, the Chi Square distribution approaches a normal distribution. Figure 1 shows density functions for three Chi Square distributions. Notice how the skew decreases as the degrees of freedom increase.

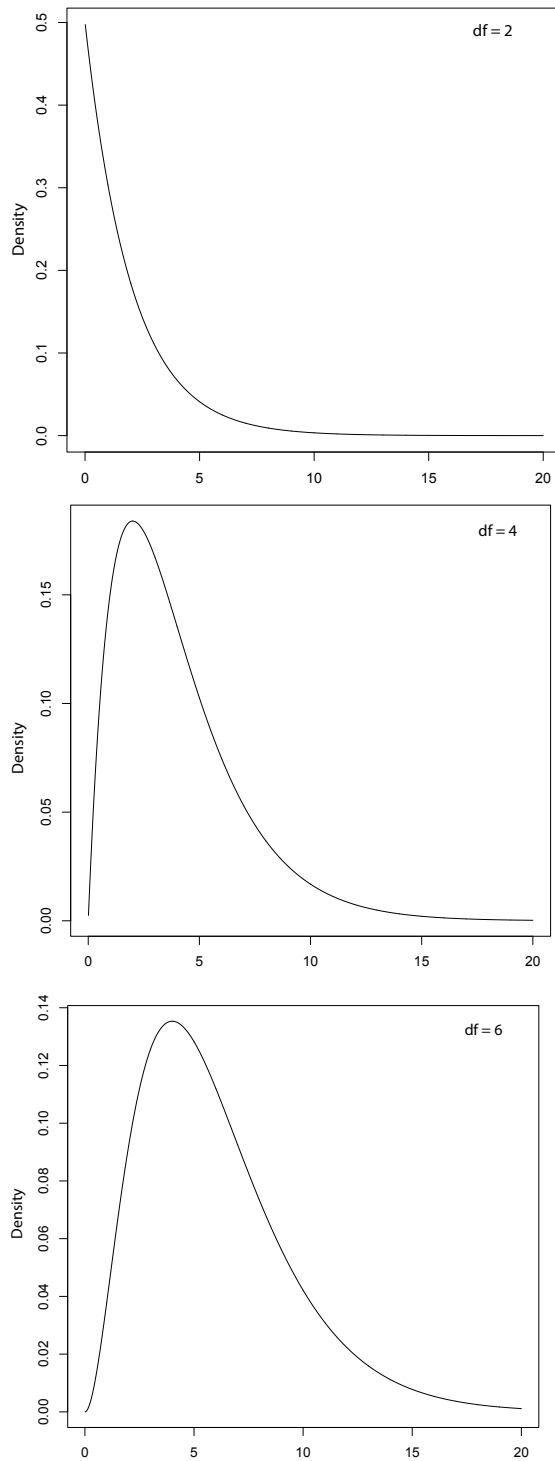


Figure 1. Chi Square distributions with 2, 4, and 6 degrees of freedom.

The Chi Square distribution is very important because many test statistics are approximately distributed as Chi Square. Two of the more common tests using the

Chi Square distribution are tests of deviations of differences between theoretically expected and observed frequencies (one-way tables) and the relationship between categorical variables (contingency tables). Numerous other tests beyond the scope of this work are based on the Chi Square distribution.

One-Way Tables (Testing Goodness of Fit)

by David M. Lane

Prerequisites

- Chapter 5: Basic Concepts of Probability
- Chapter 11: Significance Testing
- Chapter 17: Chi Square Distribution

Learning Objectives

1. Describe what it means for there to be theoretically-expected frequencies
2. Compute expected frequencies
3. Compute Chi Square
4. Determine the degrees of freedom

The Chi Square distribution can be used to test whether observed data differ significantly from theoretical expectations. For example, for a fair six-sided die, the probability of any given outcome on a single roll would be $1/6$. The data in Table 1 were obtained by rolling a six-sided die 36 times. However, as can be seen in Table 1, some outcomes occurred more frequently than others. For example, a “3” came up nine times, whereas a “4” came up only two times. Are these data consistent with the hypothesis that the die is a fair die? Naturally, we do not expect the sample frequencies of the six possible outcomes to be the same since chance differences will occur. So, the finding that the frequencies differ does not mean that the die is not fair. One way to test whether the die is fair is to conduct a significance test. The null hypothesis is that the die is fair. This hypothesis is tested by computing the probability of obtaining frequencies as discrepant or more discrepant from a uniform distribution of frequencies as obtained in the sample. If this probability is sufficiently low, then the null hypothesis that the die is fair can be rejected.

Table 1. Outcome Frequencies from a Six-Sided Die

Outcome	Frequency
1	8
2	5
3	9

4	2
5	7
6	5

The first step in conducting the significance test is to compute the expected frequency for each outcome given that the null hypothesis is true. For example, the expected frequency of a “1” is 6, since the probability of a “1” coming up is 1/6 and there were a total of 36 rolls of the die.

$$\text{Expected frequency} = (1/6) (36) = 6$$

Note that the expected frequencies are expected only in a theoretical sense. We do not really “expect” the observed frequencies to match the “expected frequencies” exactly.

The calculation continues as follows. Letting E be the expected frequency of an outcome and O be the observed frequency of that outcome, compute

$$\frac{(E - O)^2}{E}$$

for each outcome. Table 2 shows these calculations.

Table 2. Outcome Frequencies from a Six-Sided Die

Outcome	E	O	(E-O) ² /E
1	6	8	0.667
2	6	5	0.167
3	6	9	1.500
4	6	2	2.667
5	6	7	0.167
6	6	5	0.167

Next we add up all the values in Column 4 of Table 2.

$$\sum \frac{(E - O)^2}{E} = 5.33$$

This sampling distribution of

$$\sum \frac{(E - O)^2}{E}$$

is approximately distributed as Chi Square with k-1 degrees of freedom, where k is the number of categories. Therefore, for this problem the test statistic is

$$\chi^2_5 = 5.333$$

which means the value of Chi Square with 5 degrees of freedom is 5.333.

From a Chi Square calculator it can be determined that the probability of a Chi Square of 5.333 or larger is 0.377. Therefore, the null hypothesis that the die is fair cannot be rejected.

The Chi Square test can also be used to test other deviations between expected and observed frequencies. The following example shows a test of whether the variable “University GPA” in the SAT and College GPA case study is normally distributed.

The first column in Table 3 shows the normal distribution divided into five ranges. The second column shows the proportions of a normal distribution falling in the ranges specified in the first column. The expected frequencies (E) are calculated by multiplying the number of scores (105) by the proportion. The final column shows the observed number of scores in each range. It is clear that the observed frequencies vary greatly from the expected frequencies. Note that if the distribution were normal, then there would have been only about 35 scores between 0 and 1, whereas 60 were observed.

Table 3. Expected and Observed Scores for 105 University GPA Scores.

Range	Proportion	E	O
Above 1	0.159	16.695	9
0 to 1	0.341	35.805	60
-1 to 0	0.341	35.805	17
Below -1	0.159	16.695	19

The test of whether the observed scores deviate significantly from the expected scores is computed using the familiar calculation.

$$\chi^2_3 = \sum \frac{(E - O)^2}{E} = 30.09$$

The subscript “3” means there are three degrees of freedom. As before, the **degrees of freedom is the number of outcomes minus one**, which is three in this example. The Chi Square distribution calculator shows that **p < 0.001 for this Chi Square**. Therefore, **the null hypothesis that the scores are normally distributed can be rejected**.

Contingency Tables

by David M. Lane

Prerequisites

- Chapter 17: Chi Square Distribution
- Chapter 17: One-Way Tables

Learning Objectives

1. State the null hypothesis tested concerning contingency tables
2. Compute expected cell frequencies
3. Compute Chi Square and df

This section shows how to use Chi Square to test the relationship between nominal variables for significance. For example, Table 1 shows the data from the Mediterranean Diet and Health case study.

Table 1. Frequencies for Diet and Health Study

Diet	Outcome				
	Cancers	Fatal Heart Disease	Non-Fatal Heart Disease	Healthy	Total
AHA	15	24	25	239	303
Mediterranean	7	14	8	273	302
Total	22	38	33	512	605

The question is whether there is a significant relationship between diet and outcome. The first step is to compute the expected frequency for each cell based on the assumption that there is no relationship. These expected frequencies are computed from the totals as follows. We begin by computing the expected frequency for the AHA Diet/Cancers combination. Note that 22/605 subjects developed cancer. The proportion who developed cancer is therefore 0.0364. If there were no relationship between diet and outcome, then we would expect 0.0364 of those on the AHA diet to develop cancer. Since 303 subjects were on the AHA diet, we would expect $(0.0364)(303) = 11.02$ cancers on the AHA diet. Similarly, we would expect $(0.0364)(302) = 10.98$ cancers on the Mediterranean diet. In

general, the expected frequency for a cell in the i^{th} row and the j^{th} column is equal to

$$E_{i,j} = \frac{T_i T_j}{T}$$

where $E_{i,j}$ is the expected frequency for cell i,j , T_i is the total for the i^{th} row, T_j is the total for the j^{th} column, and T is the total number of observations. For the AHA Diet/Cancers cell, $i = 1$, $j = 1$, $T_i = 303$, $T_j = 22$, and $T = 605$. Table 2 shows the expected frequencies (in parenthesis) for each cell in the experiment. Table 2 shows the expected frequencies (in parenthesis) for each cell in the experiment.

Table 2. Observed and Expected Frequencies for Diet and Health Study

	Outcome				
Diet	Cancers	Fatal Heart Disease	Non-Fatal Heart Disease	Healthy	Total
AHA	15 (11.02)	24 (19.03)	25 (16.53)	239 (256.42)	303
Mediterranean	7 (10.98)	14 (18.97)	8 (16.47)	273 (255.58)	302
Total	22	38	33	512	605

The significance test is conducted by computing Chi Square as follows.

$$\chi^2_3 = \sum \frac{(E - O)^2}{E} = 16.55$$

The degrees of freedom is equal to $(r-1)(c-1)$, where r is the number of rows and c is the number of columns. For this example, the degrees of freedom is $(2-1)(4-1) = 3$. The Chi Square calculator can be used to determine that the probability value for a Chi Square of 16.55 with three degrees of freedom is equal to 0.0009. Therefore, the null hypothesis of no relationship between diet and outcome can be rejected.

A key assumption of this Chi Square test is that each subject contributes data to only one cell. Therefore, the sum of all cell frequencies in the table must be the same as the number of subjects in the experiment. Consider an experiment in

which each of 16 subjects attempted two anagram problems. The data are shown in Table 3.

Table 3. Anagram problem data

	Anagram 1	Anagram 2
Solved	10	4
Did not Solve	6	12

It would not be valid to use the Chi Square test on these data since each subject contributed data to two cells: one cell based on their performance on Anagram 1 and one cell based on their performance on Anagram 2. The total of the cell frequencies in the table is 32, but the total number of subjects is only 16.

The formula for Chi Square yields a statistic that is only approximately a Chi Square distribution. In order for the approximation to be adequate, the total number of subjects should be at least 20. Some authors claim that the correction for continuity should be used whenever an expected cell frequency is below 5. Research in statistics has shown that this practice is not advisable. For example, see: Bradley, Bradley, McGrath, & Cutcomb (1979). The correction for continuity when applied to 2 x 2 contingency tables is called the Yates correction.

Statistical Literacy

by David M. Lane

Prerequisites

- Chapter 17: Contingency Tables

An experiment was conducted to test whether the spice saffron can inhibit liver cancer. Two groups of rats were tested. Both groups were injected with chemicals known to increase the chance of liver cancer. The experimental group was fed saffron ($n = 24$) whereas the control group was not ($n = 8$). The experiment is [described here](#).

Only 4 of the 24 subjects in the saffron group developed cancer as compared to 6 of the 8 subjects in the control group.

What do you think?

What method could be used to test whether this difference between the experimental and control groups is statistically significant?

The Chi Square test of contingency tables could be used. It yields a Chi Squared ($df = 1$) of 9.50 which has an associated p of 0.002. □

References

Bradley, D. R., Bradley, T. D., McGrath, S. G., & Cutcomb, S. D. (1979) Type I error rate of the chi square test of independence in $r \times c$ tables that have small expected frequencies. *Psychological Bulletin*, 86, 1200-1297.